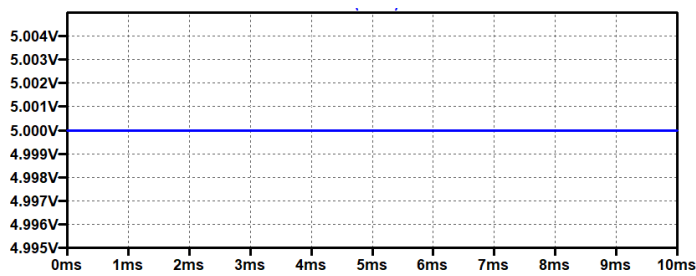


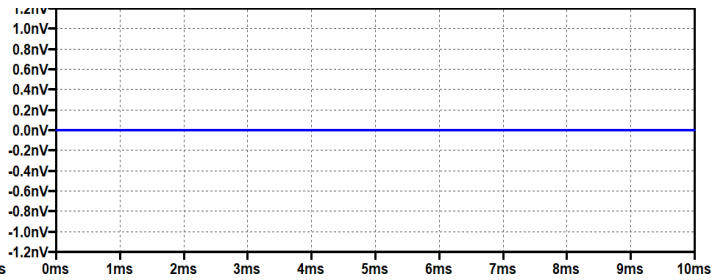
Simulation Results:

Tilt Sensor outputs:

When idle:

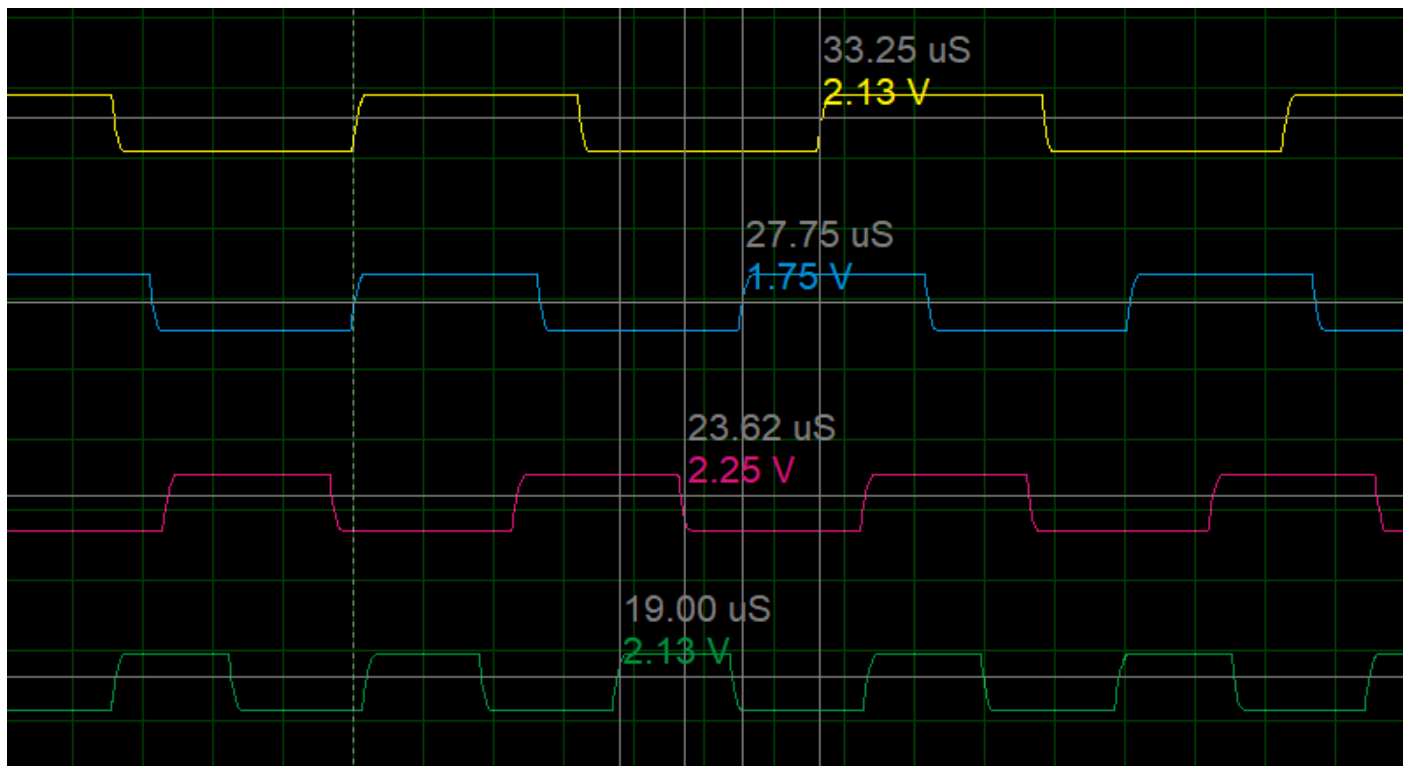


When tilted:



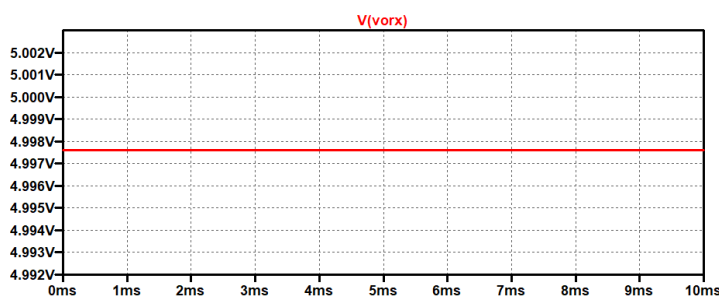
Transmitter outputs:

Frequencies: 30k Hz, 36k Hz, 40k Hz and 56k Hz, one to transmit each tilt sensor output

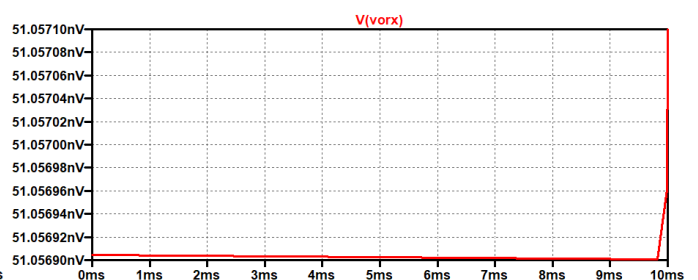


Receiver outputs:

When tilt detected:

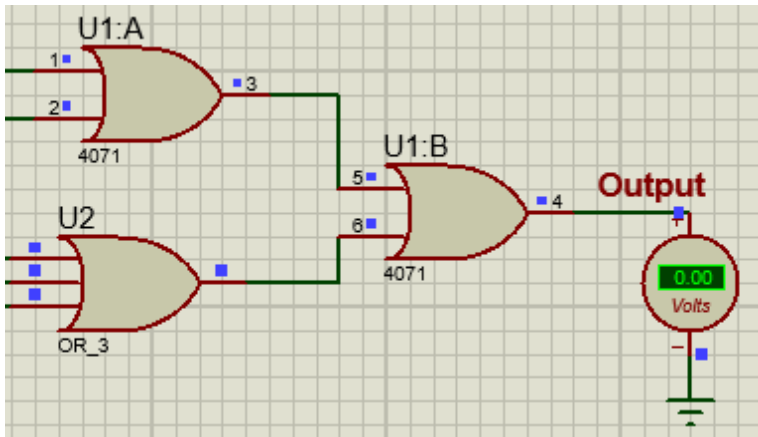


When no tilt detected:

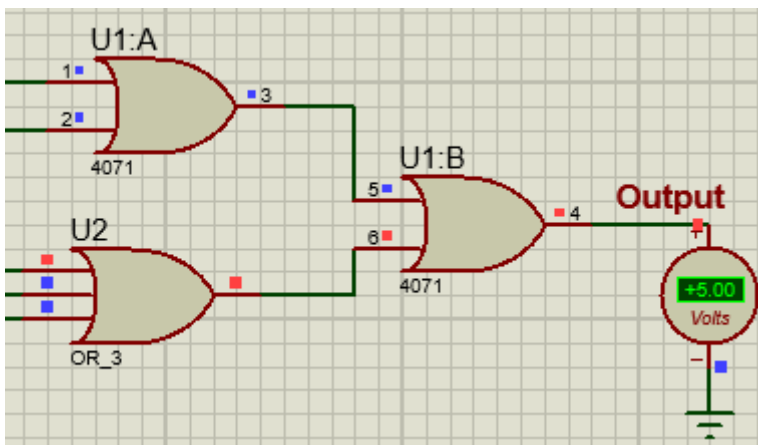


Obstacle Detector Outputs:

When no obstacle is detected:

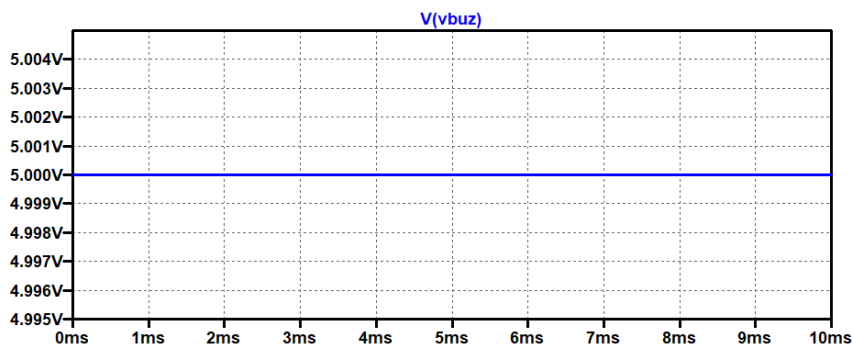


When any of the obstacle is detected:

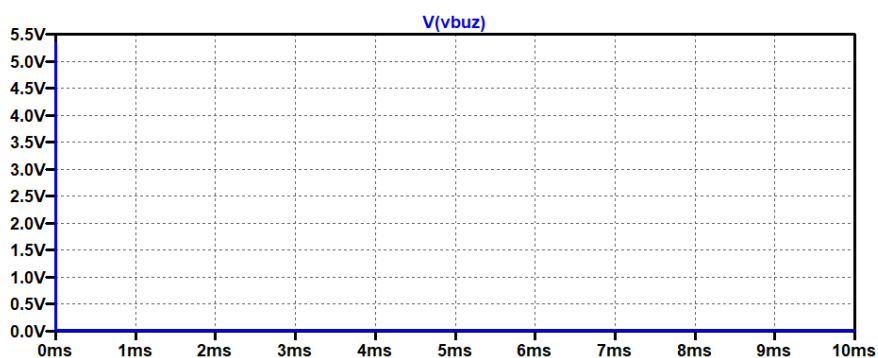


Fall Detector output:

When tilt/fall is detected:

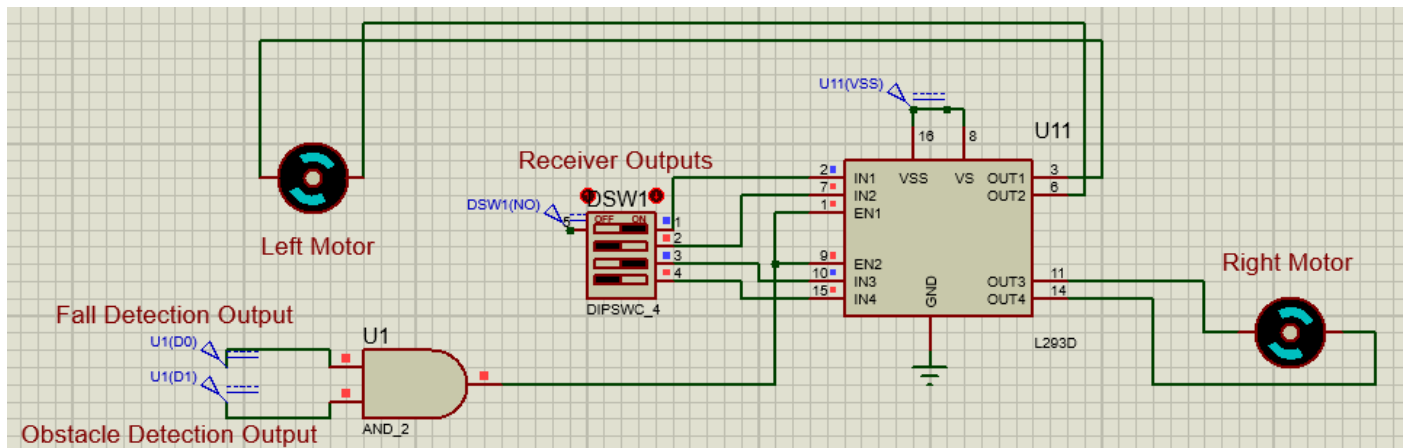


When tilt/fall is not detected:



Motor Driver Circuit output:

When the user input is detected and the direction of the motor is controlled: Motor rotates



When any of obstacle/fall is detected: Motor is idle

